KRISHISYNC AI | VIBE CODING SERIES | 3 OF 5

UI/UX Design System

Farmer-first screens, tokens and the prompts to build them

Market-Aware Agri-Coordination & Smart Energy Optimization for Indian Smallholders

Challenge 1: Sustainable Agriculture & Resource Optimization

UI/UX Design System

The user is a smallholder farmer on a low-cost Android phone, often with weak connectivity and limited literacy. Every design decision below follows from that.

Design principles

- **Icon first, text second.** Every action has a clear icon; labels are short and translated.
- **Colour carries meaning.** Green means safe, amber means caution, red means act now. Never rely on colour alone: pair with an icon and a word.
- **One decision per screen.** Show the single most important thing at the top.
- **Large touch targets.** Minimum 48 px, generous spacing, no tiny links.
- **Works on bad networks.** PWA, cached last-known data, visible 'last updated' time.
- **Multilingual from day one.** No hard-coded strings; everything is an i18n key. Hindi and English first.

Visual tokens (proposed)

Token	Value	Use
Primary green	#2E7D32	Primary buttons, safe status
Amber	#F9A825	Caution status, warnings
Alert red	#C62828	Glut warning, pump fault
Soil brown	#6D4C41	Soil and terrain elements
Background	#F6F8F3	Page background
Font	Noto Sans + Noto Sans Devanagari	Covers Hindi and other Indian scripts
Base size	16 px body, 20 px key numbers	Readable outdoors

Screen inventory

Screen	Purpose	Key components
1 Language and login	Pick language, phone OTP	Language tiles, OTP input
2 Land onboarding	Register location and terrain	Map pin, terrain picker with icons, soil type, area, pump type
3 Home dashboard	Today at a glance	Glut status banner, next pump run, weather chip, soil moisture gauge
4 Market check	Anti-glut view and recommendations	Radius heat-map, saturation bars per crop, ranked alternatives with reasons
5 Dynamic calendar	Seeding, watering, nutrient timeline	Week strip, task cards, rain-pause badges
6 Soil and advisory	Moisture and N-P-K	Gauges, trend sparkline, plain-language advice
7 Energy panel	Smart grid sync	Off-peak share, schedule timeline, idle-load alert, solar window
8 Alerts	All warnings	Severity-sorted list, tap for detail

Glut warning pattern

The glut banner is the signature interaction. It must be understood in two seconds.

State	Saturation	Banner	Message (en)
Safe	below 60%	Green, check icon	Good market room for this crop nearby.
Caution	60% to 85%	Amber, warning icon	Many farms near you plan this crop. Consider an alternative.
Glut risk	85% and above	Red, stop icon	Too much of this crop is planned within 20 km. Prices may fall.

Thresholds are defaults from the Algorithms PDF and are configurable.

Accessibility checklist

- Contrast ratio at least 4.5:1 for text.
- Status never conveyed by colour alone.
- Numbers have units and a one-line plain-language meaning.
- Phase 2: voice read-out of alerts in the farmer's language (the deck's vernacular voice interface).

UI/UX Vibe Prompts

Paste the Master Prompt (PDF 1) first. Backend can be mocked: ask the assistant to start with a typed mock API layer.

VIBE PROMPT 1 | Frontend scaffold and i18n

```
Create a Vite + React + TypeScript PWA with react-router, react-i18next, Recharts and
react-leaflet. Add Hindi (hi) and English (en) locale files and a language switcher.
Rule: no user-visible string may be hard-coded; everything uses t('key'). Add a design
tokens file (colors, spacing, font sizes) from the UI/UX PDF, Noto Sans + Noto Sans
Devanagari, 16px base, 48px minimum touch targets. Create src/api/client.ts with a typed
mock layer I can later swap for the real FastAPI base URL. Show me the folder tree when done.
```

VIBE PROMPT 2 | Onboarding flow

```
Build screens 1 and 2: language tiles, phone OTP (mock verification), then land
registration with a Leaflet map pin, an icon-based terrain picker (plain, sloped, valley,
river-side), soil type, area in hectares and pump type (grid / solar / diesel). Validate
inputs, show friendly errors in the selected language, and POST to /farms through the API
client.
```

VIBE PROMPT 3 | Home dashboard

```
Build the Home dashboard: a large glut status banner (green/amber/red with icon and text),
a card for the next scheduled pump run, a weather chip with rain chance, and a soil
moisture gauge. Each card is tappable to its detail screen. Show 'updated X minutes ago'
and keep working from cached data when offline. Use loading skeletons, not spinners.
```

VIBE PROMPT 4 | Market check and glut heat-map

```
Build the Market Check screen: a Leaflet map with a 20 km radius circle around the farm
and a heat layer of aggregated crop density (no individual farmer markers). Below it, a
list of crops with a saturation bar coloured by band and, for the top 3 alternatives, a
short reason such as 'low local supply, strong price trend'. Data shape comes from
GET /farms/{id}/glut-map and /recommendations; mock both first.
```

VIBE PROMPT 5 | Calendar, advisory and energy panel

```
Build the Dynamic Calendar (week strip of task cards: sow, irrigate, fertilise; rain-pause
badge when a pump run is skipped), the Soil screen (moisture and N-P-K gauges with a 7-day
sparkline and one plain-language sentence of advice), and the Energy panel (24-hour
timeline showing peak vs off-peak vs solar windows, scheduled pump runs overlaid, an
off-peak percentage, and an amber alert if idle power draw is flagged).
```

VIBE PROMPT 6 | Polish and demo mode

```
Add a hidden demo toggle (long-press the logo) that switches the app to a scripted scenario:
glut banner turns amber then red as a crop's saturation rises, a rain forecast pauses a
pump run, and an idle-load alert appears. Add an empty state and an error state for every
screen. Run a Lighthouse check and fix the top PWA and accessibility issues.
```